# Evaluating LLM Performance on Discipline-Specific Exams

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# 1 Introduction

Large Language Models (LLMs) are increasingly being integrated into modern educational environments, where they serve as tools for learning support, problem-solving, and exam preparation. Students utilize these models to understand complex concepts, generate explanations, and practice exam-style questions, while educators leverage them for content creation, feedback generation, and assessment assistance. This growing reliance highlights the transformative role of LLMs in reshaping how knowledge is accessed and evaluated in academic settings.

Despite their widespread adoption, a fundamental concern remains unresolved: do LLMs genuinely understand the underlying concepts they process, or do they primarily rely on pattern recognition and memorization derived from their training data? While LLMs often produce fluent and contextually appropriate responses, it is unclear whether these outputs reflect true reasoning and comprehension or simply the reproduction of learned statistical patterns.

Evaluating this distinction is particularly important in the context of exam-style questions, which are designed to assess not only factual recall but also conceptual understanding and analytical reasoning. Reliable performance in such settings is essential for determining whether LLMs can be trusted as educational tools. Moreover, understanding their limitations is critical for designing fair assessment systems that minimize the risk of AI-assisted academic misconduct.

Most existing evaluations of LLMs are based on general-purpose benchmarks, which often fail to capture the complexity, structure, and discipline-specific nature of real academic exams. These benchmarks typically emphasize broad knowledge coverage but do not adequately reflect the cognitive demands of structured examination environments.

In this report, we address this gap by evaluating LLM performance on standardized multidisciplinary exam questions. Our study systematically analyzes model accuracy, variation in performance across different levels of reasoning complexity, and discipline-specific error patterns. Through this approach, we aim to provide a more realistic and comprehensive understanding of LLM capabilities and limitations in academic assessment contexts.

# 2 Dataset Description

We used a dataset based on the OpenBookQA format, adapted to represent multiple academic disciplines and exam-style conditions. The dataset is designed to simulate real-world examination settings, where questions require varying levels of cognitive effort, ranging from simple factual recall to complex analytical reasoning. By structuring the dataset in this manner, we aim to create a realistic evaluation environment for assessing the capabilities of Large Language Models (LLMs).

## 2.1 Dataset Characteristics

The dataset includes the following key characteristics:

- **Multiple-choice questions (A/B/C/D):** Each question is presented with four possible answer options, ensuring a standardized evaluation format.

- **Exam-style structure:** Questions are formulated to resemble real academic exams, requiring precise understanding and selection of the correct answer.
- **Objective automatic scoring:** Since each question has a clearly defined correct answer, model responses can be evaluated automatically without human bias.
- **Focus on memory, conceptual, and analytical reasoning:** The dataset is intentionally designed to cover different levels of cognitive complexity, enabling a comprehensive analysis of model performance.

Each discipline contains 50 questions, ensuring a balanced distribution across subject areas. This uniform structure allows for fair comparison across different models and minimizes bias introduced by unequal data representation.

## 2.2 Reasoning Complexity Annotation

To better understand how reasoning difficulty affects model performance, each question was manually categorized into three levels of complexity:

- **Easy (Memory-based):** These questions require direct recall of factual knowledge. They typically involve retrieving known information without requiring deeper reasoning or interpretation.
- **Medium (Conceptual):** These questions require understanding and applying concepts. Models must interpret the question, relate it to the underlying principles, and select the correct answer based on conceptual knowledge.
- **Hard (Analytical):** These questions involve multi-step reasoning, logical inference, and causal analysis. They often require combining multiple pieces of information and performing structured reasoning to arrive at the correct answer.

This labeling enables analysis of how reasoning difficulty impacts model performance.

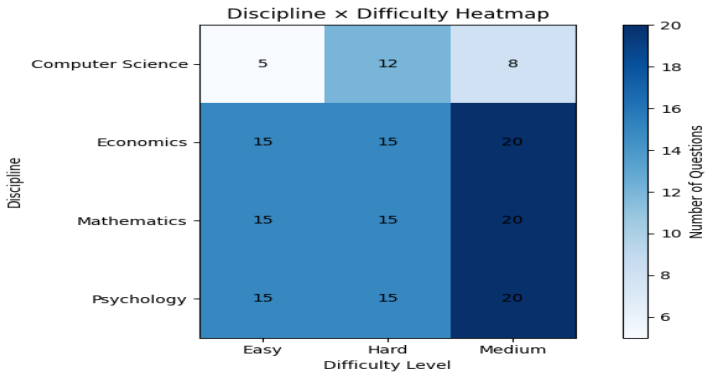


Figure 1: Overview of the Dataset Structure and Distribution

This labeling enables a systematic analysis of how the complexity of reasoning impacts the performance of the model. By separating questions into these categories, we can observe performance trends, identify weaknesses in analytical reasoning, and better understand the limitations of LLMs in handling increasingly complex tasks.

# 3 Methodology

## 3.1 Evaluation Pipeline

To ensure a systematic and fair comparison across different Large Language Models (LLMs), we designed a structured evaluation pipeline. The pipeline standardizes the interaction between models and the dataset, ensuring that all models are evaluated under identical conditions.

The evaluation process consists of the following steps:

1. **Model Evaluation with Standardized Prompts:** Each model (GPT Go, Gemini 3, and Grok 4.1) is evaluated using identical exam-style prompts to ensure consistency and eliminate prompt-related bias.

2. **Answer Generation:** The models generate responses for each multiple-choice question in the dataset.

3. **Automatic Answer Comparison:** The generated responses are automatically compared with the ground truth answers provided in the dataset.

4. **Metric Computation:** Accuracy and error-related metrics are computed to quantify model performance.

5. **Statistical Analysis:** One-Way ANOVA is applied to determine whether observed performance differences between models are statistically significant.

This pipeline ensures reproducibility and fairness, allowing us to isolate model performance from external factors such as prompt variation or evaluation inconsistency.

## 3.2 Prompt Engineering

Prompt engineering plays a critical role in influencing LLM performance. To ensure a fair evaluation, prompt optimization was applied to reduce ambiguity and guide model behavior.

Prompt refinement helps to:

- **Reduce ambiguity:** Clear and structured prompts minimize misinterpretation of questions.

- **Guide reasoning:** Well-designed prompts encourage models to follow logical steps when generating answers.

- **Improve answer selection:** Explicit instructions help models choose the most appropriate option among multiple choices.

By refining prompts, we aim to distinguish between errors caused by unclear instructions and those resulting from limitations in model reasoning. This enables a more accurate assessment of model capabilities.

# 4 Training and Evaluation

## 4.1 Experimental Setup

The experimental setup is designed to evaluate model performance comprehensively across multiple dimensions. The following evaluation metrics are used:

- **Accuracy:** Measures the percentage of correctly predicted answers.
- **Error Percentage:** Represents the proportion of incorrect responses, providing insight into model weaknesses.
- **Difficulty-wise Performance:** Evaluates model accuracy across different levels of reasoning complexity.

To further validate the results, statistical analysis is performed using One-Way ANOVA. This test helps determine whether the differences in performance across models are statistically significant or occur due to random variation.

The hypothesis for ANOVA is defined as follows:

- **Null Hypothesis ($H_0$):** There is no significant difference between the performance of the models.
- **Alternative Hypothesis ($H_1$):** At least one model performs significantly differently from the others.

## 4.2 Difficulty-wise Evaluation

To analyze reasoning capabilities in detail, model performance is evaluated separately for different levels of question difficulty:

- **Easy Questions:** Evaluate the model's ability to recall factual information.
- **Medium Questions:** Assess conceptual understanding and the ability to apply knowledge.
- **Hard Questions:** Test analytical reasoning, multi-step problem-solving, and logical inference.

By computing accuracy for each difficulty level, we can observe how model performance degrades as cognitive complexity increases. This provides valuable insights into the strengths and limitations of LLMs, particularly in handling analytical and reasoning-intensive tasks.

# 5 Results

## 5.1 Overall Model Performance (RQ1)

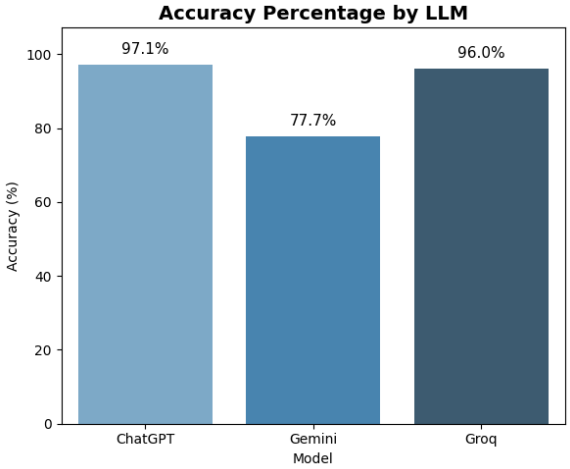


Figure 2: Before Prompt engineering

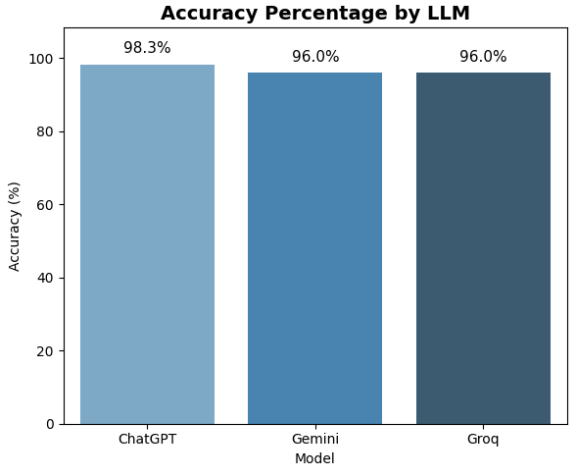


Figure 3: After Prompt engineering

After prompt optimization, the results are:

- ChatGPT GO: 98.3%
- Gemini 3: 96.0%
- Grok 4.1: 96.0%

Gemini showed the largest improvement (+18.3%).
ANOVA results:

- F-statistic: 0.97
- p-value: 0.379

This indicates that the performance differences are not statistically significant.

## 5.2 Performance by Reasoning Complexity (RQ2)

The results show that:

- Accuracy declines from Easy to Medium to Hard
- The largest performance drop occurs for Hard questions
- Easy and Medium accuracy improved after prompt refinement
- Hard questions remain challenging

| Model | ChatGPT | Gemini | Groq |
|---|---|---|---|
| Difficulty | | | |
| Easy | 100.000000 | 100.000000 | 96.000000 |
| Hard | 94.736842 | 91.228070 | 94.736842 |
| Medium | 100.000000 | 97.058824 | 97.058824 |

Figure 4: Performance by Reasoning Complexity

### 5.3 Error Analysis (RQ3)

Error patterns reveal that:

- Higher error rates occur in Computer Science and Economics
- Lower error rates occur in Psychology and factual disciplines
- Errors are systematic rather than random

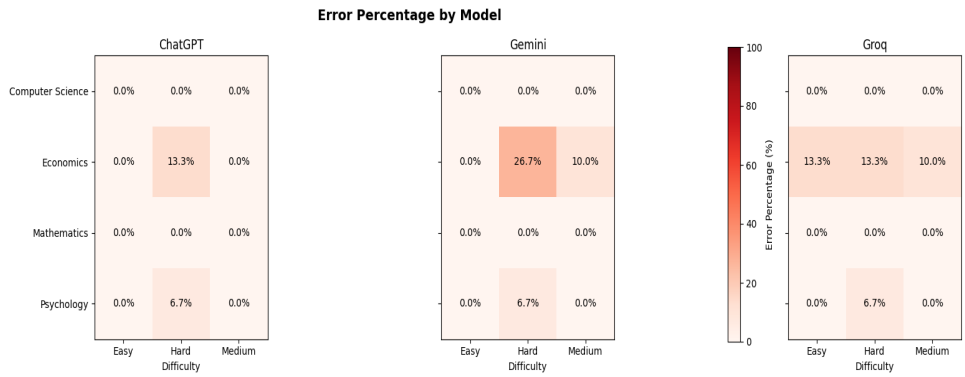


Figure 5: Error Percentage by Model

# 6 Discussion

The results of this study provide important insights into the capabilities and limitations of Large Language Models (LLMs) when evaluated on discipline-specific exam questions. The findings indicate that LLMs perform consistently well on memory-based and conceptual tasks, where the primary requirement is factual recall or understanding of well-defined concepts. This strong performance can be attributed to the extensive training data used to train these models, which enables them to retrieve and reproduce learned information effectively.

However, a significant decline in performance is observed for analytical questions that require multi-step reasoning, logical inference, and the integration of multiple concepts. This suggests that current LLMs face inherent limitations in handling complex reasoning

tasks. These limitations are consistent across all evaluated models, indicating that the issue is not model-specific but rather a broader challenge in current LLM architectures.

Prompt engineering plays a crucial role in improving model performance. By refining prompts to reduce ambiguity and provide clearer instructions, we observed noticeable improvements in accuracy, particularly for easy and medium-level questions. However, these improvements are largely limited to better interpretation of the question rather than enhanced reasoning capability. In other words, prompt engineering helps models understand what is being asked but does not fundamentally improve how they reason.

Furthermore, discipline-specific error patterns reveal that LLM performance varies significantly across subject areas. Tasks that involve algorithmic thinking, mathematical reasoning, or causal analysis—such as those found in Computer Science and Economics—tend to produce higher error rates. In contrast, disciplines that rely more on factual knowledge or descriptive understanding show comparatively lower error rates. These findings highlight that LLM limitations are systematic and predictable, rather than random.

Overall, the discussion emphasizes that while LLMs are highly capable tools for knowledge retrieval and conceptual explanation, they are not yet reliable for tasks requiring deep analytical reasoning and structured problem-solving.

# 7 Conclusion

In this report, we conducted a comprehensive evaluation of Large Language Model (LLM) performance on discipline-specific exam questions across multiple levels of reasoning complexity. The study was designed to simulate realistic academic conditions and assess how well LLMs perform in structured examination settings.

The findings demonstrate that LLMs perform reliably on memory-based and conceptual questions, achieving high accuracy in tasks that involve factual recall and basic understanding. However, performance declines significantly as the complexity of reasoning increases, particularly for analytical questions that require multi-step logic and inference.

Additionally, the analysis reveals that errors made by LLMs are systematic and discipline-dependent. Certain subject areas, especially those involving algorithmic or causal reasoning, present greater challenges for these models. This suggests that LLM limitations are not random but are influenced by the nature of the task and the type of reasoning required.

While prompt engineering improves evaluation fairness and reduces ambiguity, it does not address the fundamental limitations in reasoning capability. Therefore, relying solely on prompt optimization is insufficient for overcoming the challenges associated with analytical tasks.

These findings highlight the importance of cautious integration of LLMs in academic environments. While they can serve as effective tools for learning and support, their limitations must be carefully considered, particularly in high-stakes assessment scenarios. Human oversight and structured evaluation mechanisms remain essential to ensure reliability and fairness.

# 8 Future Work

Although this study provides valuable insights into LLM performance, several directions for future research can further enhance understanding and improve model capabilities.

One promising approach is the use of chain-of-thought prompting, which encourages models to generate intermediate reasoning steps before arriving at a final answer. This technique has the potential to improve performance on analytical tasks by promoting structured reasoning.

Another important direction is the development of hybrid neuro-symbolic systems that combine the strengths of neural networks with symbolic reasoning methods. Such approaches may help address the limitations of LLMs in logical inference and multi-step problem-solving.

In addition, incorporating human-in-the-loop verification systems can improve reliability by allowing human experts to review and validate model outputs, particularly in critical or high-stakes applications.

Finally, future evaluations should extend beyond multiple-choice questions to include open-ended and descriptive questions. This would provide a more comprehensive assessment of LLM capabilities, particularly in generating explanations, arguments, and structured responses.

By exploring these directions, future research can contribute to the development of more robust, reliable, and reasoning-capable language models suitable for educational and academic applications.

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